



Beyond the barrel: Europe's hidden energy vulnerability

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Recent supply chain stress, highlighted by European PMI data, has renewed focus on the continent's external vulnerabilities. In this note, we argue that Europe's exposure is twofold:

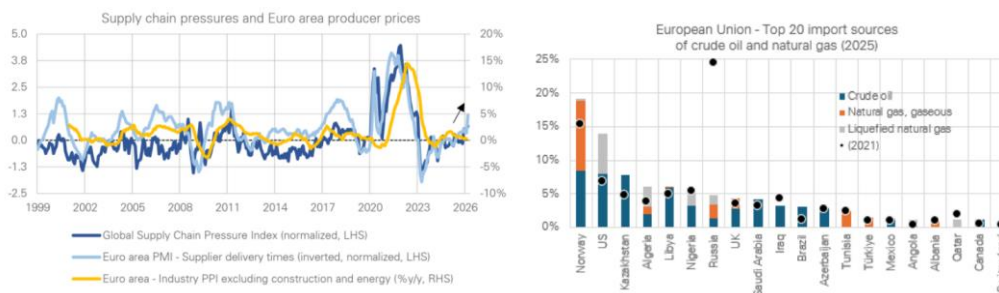
- A direct, tangible dependency on the Gulf for specific commodities like oil products (e.g., jet fuel, diesel), oil derivatives, chemicals and metals, representing a measurable cost.
- A more elusive and structural indirect risk from "hidden" energy embedded within industrial imports, particularly from Asia.

This indirect exposure is qualitatively different, comprised of refined products and electricity (not just crude oil) and threatens shortages of specific, non-fungible industrial inputs like plastics, chemicals and metals. This upstream vulnerability is far harder to substitute than direct commodity imports and, as we've argued previously, is a prime example of a non-linear risk that can amplify stagflation and complicate monetary policy.

Note: 'Direct' exposure hereafter refers to Europe's reliance on imported goods from a specific region, for instance energy from the Middle-East. 'Indirect' exposure refers only to the "embodied" content of an import – the energy consumed along the value chain to produce that good before export to Europe.

A ceasefire's reprieve, a system's vulnerability. The fragility of the agreed two-week ceasefire in the Middle East was underscored this weekend by the failure of US-Iran talks. While any pause in hostilities offers a welcome reprieve for global supply chains, this development highlights that underlying risks remain in place.

Figure 1: Supply chain pressure is increasing rapidly. EU direct energy exposure to GCC States not as pronounced as to Russia in 2022



Source: Deutsche Bank Research, NY FED, S&P Global, Eurostat (datasets nrg_ti_gas and nrg_ti_oil) for the right-hand chart. GCC countries refers to a sum of Bahrain, Iran, Kuwait, Qatar, Saudi Arabia and the UAE. 2025 trade share covers January to September.

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Recent data has already provided a clear warning signal of this vulnerability, with the March PMI releases for Europe showing a notable lengthening of supplier delivery times and a fresh acceleration in input costs (Figure 1, left).

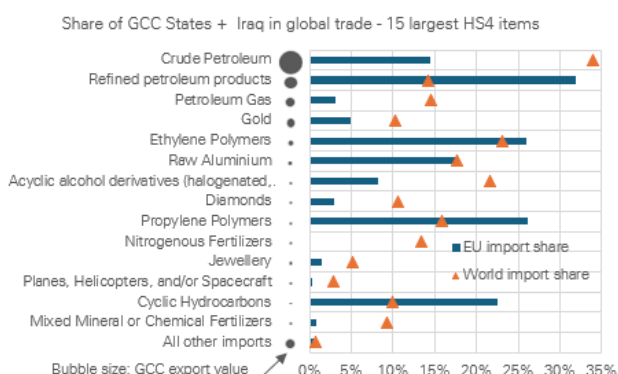
While this offers a glimpse of what could happen if the conflict escalates again, the transmission mechanism to Europe is more complex than a simple repeat of 2022. There are two distinct channels of vulnerability when it comes to this energy shock: a direct, measurable impact on the cost of specific imports – largely energy –, and a more subtle, structural and indirect risk embedded within global value chains, potentially impacting the vulnerability of various inputs.

Channel 1: A tangible, product-level exposure. The direct risk channel has two layers. The first is the overall cost of energy. As we quantified in a recent note, the euro area's energy import bill should increase by around 0.8% of GDP, based on current market pricing (Brent at 98 USD/bbl and gas at 47 EUR/MWh).

The second risk is a more specific dependency on the region central to the crisis in terms of physical supply. This is most visible not at the continent-wide level – where suppliers like Norway, the US, and Kazakhstan now dominate Europe's energy supply (Figure 1, right) – but at the product-specific level, where Gulf countries can provide upwards of 15% of EU imports. A key example is refined products, where Europe sources a notable share from the region (Figure 2). While Europe's significant domestic refining capacity (supplying 81% of the domestic market in refined petroleum in 2022) provides a crucial buffer, the import dependency remains.

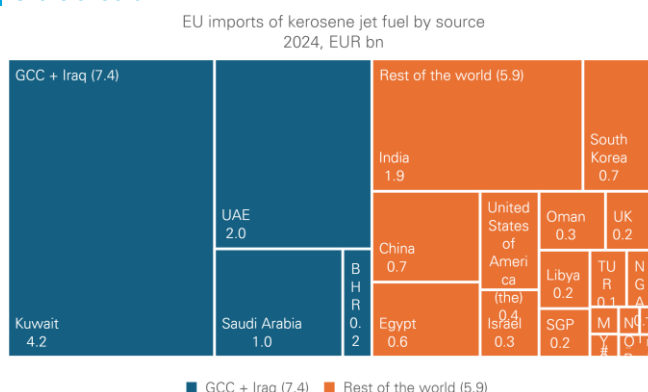
For certain types of refined petroleum products, such as kerosene, the share of GCC countries as a source for EU imports reaches 56% (Figure 3). On this point specifically, our in-house specialists think that the air transport sector has 45-55 days of effective cover, higher than the normal 30-35 days thanks to the release of reserves by the International Energy Agency. Uncertainty prevails, however, as the ceasefire has done little to ease the Strait of Hormuz situation.

Figure 2: The EU's reliance on the Gulf is meaningful in specific categories, although generally lower than the rest of the world



Source: Deutsche Bank Research, BACI

Figure 3: An example of high exposure is kerosene, with an import share of 56%



Source: Deutsche Bank Research, Eurostat

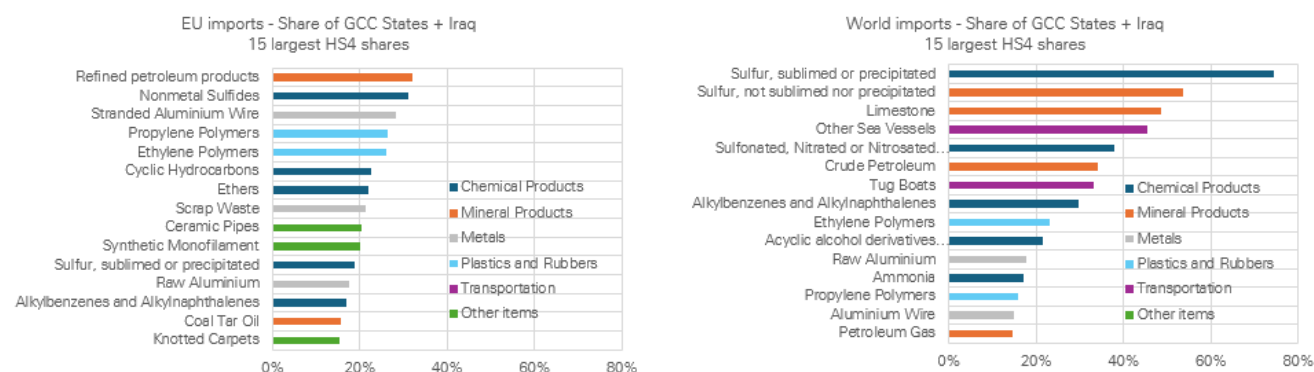
Furthermore, granular data reveals pockets of high dependency for specific petrochemicals, plastics, and metals (Figure 4, left). These concentrated



exposures represent a measurable cost to Europe's import bill and a vulnerability for the industries that rely on them. In short, while Europe's product-level exposure may be lower than in other parts of the world, it is real and cannot be dismissed. The second channel of risk, however, is qualitatively different and arguably more complex.

Channel 2: A hidden, structural risk in value chains. While Europe's visible exposure is specific, the GCC and Iraq region's role in supply chains more globally creates a broader bottleneck risk. The region's global importance extends beyond crude oil, and it holds a commanding position in key industrial inputs, particularly in mineral and chemical products (Figure 4, right).

Figure 4: The granular set of products where the Gulf makes a large fraction of EU and world imports



Source: Deutsche Bank Research, BACI

For instance, the region accounts for over 70% of global exports of certain forms of sulfur and 48% for limestone. This is complemented by significant global shares in chemical feedstocks like sulfonated/nitrated derivatives (38%) and plastics like ethylene polymers (23%).

This global bottleneck risk is one indirect threat to Europe. Another transmission channel is through second-round effects via Asia, whose industries are more reliant on these Gulf supplies. Recent headlines (see [here](#), [here](#) and [here](#)) already flag this, with reports of Asian producers cutting output due to soaring feedstock costs.

We quantify this indirect channel by calculating the energy "embodied" in Europe's imports – that is, the energy consumed all along the global value chain to produce imported goods. Based on OECD Input-Output tables for 2022 – the last year available –, we find that in addition to Europe's USD 567bn (3.8% of EU GDP) in direct energy imports, the goods imported into Europe for use in supply chains required an additional USD 370bn (2.5% of GDP) of indirect energy to be consumed in their production (Figure 5, left). A significant share of this – USD 170bn (1.1% of GDP) – was embedded in goods and services from Asia alone. These measures capture total energy use across production stages, including energy used within the energy sector itself, and is therefore not 100% comparable to observed import flows.

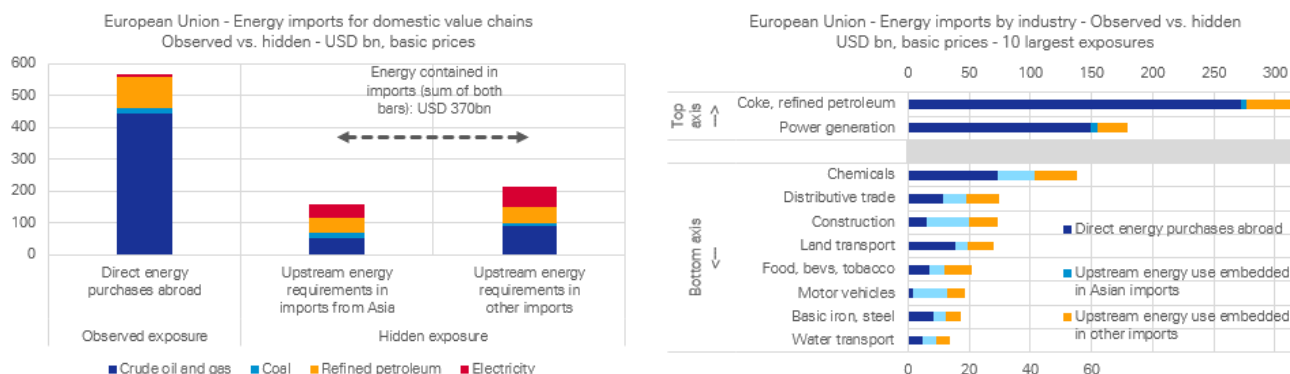
Crucially, this hidden exposure has a different risk profile. While direct imports are dominated by crude oil, the indirect energy footprint has a larger share of



refined products and electricity. This shifts the focus of risk away from just oil wells and towards the stability of foreign refining capacity and power grids – critical, yet less monitored, factors for European industrial resilience.

Pinpointing upstream vulnerabilities. What inputs could create disruptions? The impact of this indirect exposure is not uniform across the European economy. Beyond direct energy consumers (refined petroleum and power generation) sectors such as chemicals, manufacturing, and construction carry a significant indirect energy footprint tied to their Asian supply chains (Figure 5, right).

Figure 5: Energy dependency – What is observed and what hides in imports. The exposure of many manufacturing sub-segments is largely indirect



Source: Deutsche Bank Research, OECD Inter-Country Input-Output Table (2022). Analysis covers imported energy only. The OECD's \$567bn figure differs from Eurostat's ~\$700bn primarily due to valuation (OECD uses 'basic prices', Eurostat uses 'CIF') and scope (OECD tracks by exporting industry, not purely by product). The \$567bn measure also only accounts for energy as an input (intermediate demand).

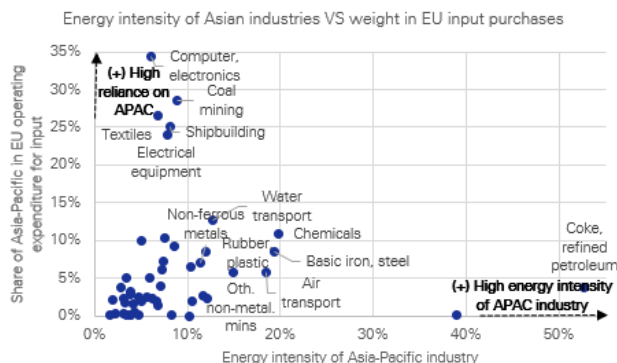
The transmission risk is most acute for inputs that combine two characteristics: high sourcing from Asia and high energy intensity. This makes basic chemicals and industrial materials – like plastics and steel – primary vectors through which cost and availability shocks can propagate into European industries (Figure 6).

Having measured total energy requirements along value chains, we now isolate the pure energy cost absorbed by each economy. This strips out non-energy components of the energy sector's output (intermediate inputs as well as margins, taxes and labour costs), providing a cleaner measure of underlying energy exposure. This shifts the focus from production requirements to income absorption/value added contributions from energy as an input.

This country analysis reveals a remarkably widespread "hidden" burden. Factoring in this absorbed energy cost adds an amount equivalent to between 0.8% and 1.5% of GDP to the import bill for most major European economies. This effectively increases their total energy dependency by a third to a half, compounding already high exposure in countries like Italy, Spain, and Poland (Figure 7).

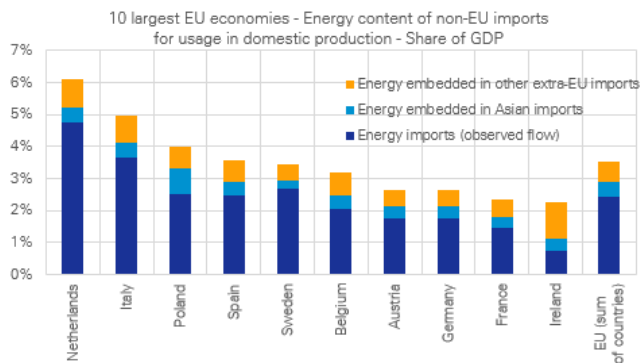


Figure 6: Which inputs from Asia could be at risk in terms of supply?



Source: Deutsche Bank Research, OECD ICIO

Figure 7: Energy absorption shifting higher when hidden energy is captured



Source: Deutsche Bank Research, OECD ICIO

Conclusion: A structural vulnerability. The key takeaway is that Europe's exposure in terms of supply is twofold. The direct part (trade exposure to Gulf countries on crude oil, gas, oil-derivatives, chemicals and metals) is tangible and should not be overlooked. The indirect part is more elusive, structural and difficult to manage: energy that arrives embedded within industrial inputs – with a precise fit in those chains – is much harder to substitute than fungible commodities like oil or natural gas, for which Europe's sourcing has already shifted to other suppliers.

This creates the potential for shortages of specific, critical components, a prime example of the non-linear risks we highlighted in a recent piece. The nature of such a structural vulnerability means it is often less responsive to broad macroeconomic tools and requires a more targeted level of monitoring. The ultimate message is that even countries with limited direct dependence to the Gulf face a meaningful – and arguably less manageable – upstream vulnerability through their reliance on global industrial supply chains.

The lingering risk of shortages amplifies the stagflation risks already prevalent from the energy shock (Figure 6), and complicates the conduct of monetary policy. Our view is that the risk of persistent inflation will likely drive the ECB to hike rates to 2.50%, the upper end of the range of neutral rates. This would signal the ECB's commitment to price stability without overburdening growth when there are downside risks.



Appendix 1

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